

# 第十七届丘成桐大学生数学竞赛（数学物理赛道）

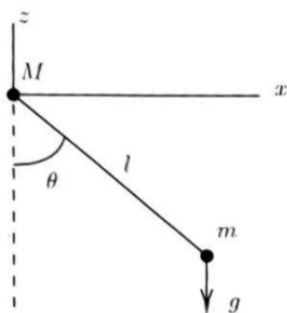
## 个人解答

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### 1 Problem 1

Consider a sliding pendulum, consisting of a mass  $M$  which can move without friction along a horizontal bar, and which is connected by a massless rod of length  $l$  to another mass  $m$ .



1. Derive the angular frequency of small oscillations in the  $x$ - $z$  plane.
2. Next consider the sliding conical (also called spherical) pendulum, for which the mass  $M$  can move without friction in the horizontal  $x$ - $y$  plane. Consider circular motion of both masses for a fixed, not necessarily small, angle  $\theta$ . What is the angular frequency as a function of  $\theta$ ? Check your result for small  $\theta$ .
3. Now consider the inverted sliding pendulum (which can only move in  $x$ - $z$  plane, i.e. back to the set-up of 1). As it starts falling from rest at  $\theta = \pi$ , there comes a point where the tension  $T$  in the rod becomes zero. Find the value of  $\theta$  when this happens, assuming for simplicity that  $M$  is infinitely large.
4. Same set-up as 3. What is the tension at  $\theta = \frac{\pi}{2}$  and  $\theta = \pi$  when the inverted sliding pendulum is falling from rest at  $\theta = \pi$  but this time  $M$  and  $m$  are arbitrary?

**Hint:** For an ellipse  $x^2/a^2 + y^2/b^2 = 1$ , the curvature radius at  $(a, 0)$  is  $b^2/a$  and at  $(0, b)$  is  $a^2/b$ .

**解答**

1. 取水平位移  $X$  和摆角  $\theta$ .  $m$  的坐标为  $x_m = X + l \sin \theta$ ,  $z_m = -l \cos \theta$ . 动能  $T = \frac{1}{2} M \dot{X}^2 + \frac{1}{2} m [(\dot{X} + l \dot{\theta} \cos \theta)^2 + (l \dot{\theta} \sin \theta)^2] = \frac{1}{2} (M + m) \dot{X}^2 + \frac{1}{2} m l^2 \dot{\theta}^2 + m l \dot{X} \dot{\theta} \cos \theta$ , 势能  $V = -mgl \cos \theta$ . 小角度近似  $\cos \theta \approx 1 - \theta^2/2$ , 保留二阶得  $T \approx \frac{1}{2} (M + m) \dot{X}^2 + \frac{1}{2} m l^2 \dot{\theta}^2 + m l \dot{X} \dot{\theta}$ ,  $V \approx -mgl(1 - \theta^2/2)$ . 拉格朗日方程给出  $(M + m) \dot{X} + m l \dot{\theta} = \text{const.}$  初始静止, 故  $\dot{X} = -\frac{m l}{M + m} \dot{\theta}$ ; 代入  $\theta$  方程得  $\frac{M m l^2}{M + m} \ddot{\theta} = -mgl \theta$ ,

即  $\ddot{\theta} + \frac{M+m}{M} \frac{g}{l} \theta = 0$ , 角频率  $\omega = \sqrt{\frac{M+m}{M} \frac{g}{l}}$ .

2. 系统绕竖直轴以角速度  $\Omega$  转动,  $m$  与竖直方向夹角  $\theta$  固定. 设  $M$  到转轴距离为  $R$ , 质心水平静止得  $MR + m(R + l \sin \theta) = 0$  即  $R = -\frac{ml \sin \theta}{M+m}$ ,  $m$  的圆周半径  $r_m = R + l \sin \theta = \frac{Ml \sin \theta}{M+m}$ . 竖直方向

$T \cos \theta = mg$ , 水平方向  $T \sin \theta = m\Omega^2 r_m$ . 消去  $T$  得  $\tan \theta = \frac{\Omega^2}{g} \cdot \frac{Ml \sin \theta}{M+m}$ , 故  $\Omega^2 = \frac{M+m}{M} \frac{g}{l} \cos \theta$ ,

即  $\omega(\theta) = \sqrt{\frac{M+m}{M} \frac{g}{l} \cos \theta}$ . 小  $\theta$  时  $\cos \theta \approx 1$  回到第 1 问结果.

3. 能量守恒给出

$$\dot{\theta}^2 = \frac{2g}{l}(1 + \cos \theta)$$

张力  $T = m(g \cos \theta + l\dot{\theta}^2)$ . 令  $T = 0$ , 解得

$$\cos \theta_0 = -\frac{2}{3}, \quad \theta_0 = \arccos(-2/3)$$

当  $\theta > \theta_0$  时杆张力变为 0.

4. 由动量守恒

$$\dot{X} = -\frac{ml \cos \theta}{M+m} \dot{\phi}$$

将其代入张力表达式

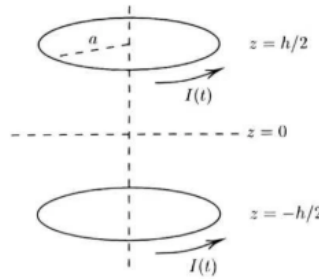
$$T = m(g \cos \theta + l\dot{\theta}^2 - \sin \theta \dot{X} \dot{\theta})$$

取初始条件  $\theta(0) = \pi$ ,  $\dot{\theta}(0) = 0$ , 计算可得, 在  $\theta = \pi$  时  $T = mg$ , 在  $\theta = \pi/2$  时

$$T\left(\frac{\pi}{2}\right) = \frac{2Mmg}{M+m}$$

## 2 Problem 2

Two identical circular coils of wire of radius  $a$  are separated by a length  $h$ . The coils each carry a slowly varying sinusoidal current  $I(t) = I_0 \cos(\omega t)$ . The axis of the coils is aligned with the  $z$ -axis and the geometry is centered at  $z = 0$ . (See below)



1. At lowest order in the frequency, a magnetostatic approximation is valid. Using this approximation, show that close to the axis, and near  $z = 0$ , the Taylor series for the axial and radial components of the slightly off-axis magnetic field take the approximate form:

$$B_z \approx B_0 + \frac{1}{2}\beta_2 \left( z^2 - \frac{\rho^2}{2} \right) + \dots$$

$$B_\rho \approx -\frac{1}{2}\beta_2 z \rho + \dots$$

where  $B_0$  and  $\beta_2$  are determined by the Taylor series of the magnetic field on the z-axis:

$$B_z(z) \approx B_0 + \frac{1}{2}\beta_2 z^2 + \dots$$

Here  $\rho = \sqrt{x^2 + y^2}$ .

2. Using the magnetostatic approximation, determine the magnetic field in the z direction close to the axis of the solenoid, and near  $z = 0$ , to quadratic order in  $z$  and  $\rho$ . Describe the magnetic field when  $h = a$ .
3. Determine the electric field close to the axis of the solenoid at  $z = 0$  to the lowest non-trivial order in the frequency and  $\rho$ .
4. Briefly answer the following:
  - (a) For parts 2 and 3, give an estimate for the size of finite-frequency corrections.
  - (b) For part 2, estimate at what large  $z$  the magnetostatic approximation breaks down when computing the magnetic field on the z-axis.

**解答**

1. 在无电流的区域内  $\nabla \times \mathbf{B} = 0$  且  $\nabla \cdot \mathbf{B} = 0$ . 可引入磁标势  $\Phi$  使

$$\mathbf{B} = -\nabla\Phi, \quad \nabla^2\Phi = 0$$

柱坐标  $(\rho, \phi, z)$  中

$$\Phi(z, \rho) = \phi_0(z) + \phi_2(z)\rho^2 + \phi_4(z)\rho^4 + \dots$$

把它代入拉普拉斯方程并由同阶  $\rho$  的系数相等得到

$$\phi_2 = -\frac{1}{4}\phi_0'', \quad \phi_4 = -\frac{1}{16}\phi_2'', \dots$$

于是

$$B_z = -\partial_z\Phi = -\phi_0' - \phi_2'\rho^2 - \dots = B_0(z) - \frac{B_0''(z)}{4}\rho^2 + \dots$$

$$B_\rho = -\partial_\rho\Phi = -2\phi_2\rho - \dots = -\frac{B_0'(z)}{2}\rho + \dots$$

其中  $B_0(z) \equiv B_z(z, 0) \equiv -\phi_0'(z)$

对  $z = 0$  的泰勒展开

$$B_0(z) = B_0 + \frac{1}{2}\beta_2 z^2 + \mathcal{O}(z^4), \quad B_0''(0) = \beta_2.$$

把上式代回即得到

$$B_z(\rho, z) \approx B_0 + \frac{\beta_2}{2}(z^2 - \frac{1}{2}\rho^2)$$

$$B_\rho(\rho, z) \approx -\frac{\beta_2}{2}z\rho$$

这正是题目中给出的近似形式.

2. 单条半径为  $a$  的闭合线圈在轴上距其中心的距离  $z$  处的磁场

$$B_l(z) = \frac{\mu_0 I a^2}{2(a^2 + z^2)^{3/2}}$$

两条线圈分别位于  $z = \pm h/2$ , 于是

$$B_z(z) = \frac{\mu_0 I a^2}{2} \left[ \frac{1}{(a^2 + (z - h/2)^2)^{3/2}} + \frac{1}{(a^2 + (z + h/2)^2)^{3/2}} \right]$$

在  $z = 0$  处

$$B_0 = \frac{\mu_0 I a^2}{(a^2 + (h/2)^2)^{3/2}}$$

取二阶导数得到

$$\beta_2 = -\frac{3\mu_0 I a^2 (a^2 - h^2)}{(a^2 + (h/2)^2)^{7/2}}$$

给出在  $z = 0$  附近、 $\rho$  乘到二次项的磁场

$$B_z(\rho, z) \approx B_0 + \frac{\beta_2}{2} (z^2 - \frac{1}{2}\rho^2), \quad B_\rho(\rho, z) \approx -\frac{\beta_2}{2} z\rho$$

当  $h = a$  时

$$B_0 = \frac{\mu_0 I a^2}{(a^2 + (a/2)^2)^{3/2}} = \frac{8\mu_0 I}{5\sqrt{5} a}$$

而

$$\beta_2 = -\frac{3\mu_0 I a^2 (a^2 - h^2)}{(a^2 + (h/2)^2)^{7/2}} = 0$$

于是

$$B_z(\rho, z) \approx B_0 + \mathcal{O}(\rho^4, z^4), \quad B_\rho(\rho, z) \approx 0$$

结论：线圈间距等于半径时， $z = 0$ ， $\rho \ll a$  的局域磁场几乎完全均匀，没有二阶磁梯度，可以把它近似为理想等压场。

### 3. 由法拉第电磁感应律

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

在轴对称、静磁场近似下  $B_z(\rho, z, t) = B_z(\rho, z) \cos \omega t$ . 有

$$(\nabla \times \mathbf{E})_z = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho E_\phi) = -\frac{\partial B_z}{\partial t} = \omega B_z(\rho, z) \sin \omega t$$

令  $z = 0$ ，并把  $B_z(\rho, 0) = B_0 - \frac{\beta_2}{4} \rho^2$  代入

$$\frac{1}{\rho} \frac{d}{d\rho} (\rho E_\phi) = \omega \left( B_0 - \frac{\beta_2}{4} \rho^2 \right) \sin \omega t$$

积分得到

$$E_\phi(\rho, 0, t) = \frac{\omega B_0}{2} \rho \sin \omega t - \frac{\omega \beta_2}{16} \rho^3 \sin \omega t + \mathcal{O}(\rho^5).$$

最低非平凡阶

$$\mathbf{E}(\rho, 0, t) \approx \frac{\omega B_0}{2} \rho \sin \omega t \hat{\phi}$$

在  $\rho \rightarrow 0$  时， $E_\rho, E_z$  在此阶都为零。这正是由变磁通在非常小半径处产生的环形电场。

### 4. (a) 在低频情形下，磁场与电场的修正可用求解 Helmholtz 方程

$$(\nabla^2 + k^2) \mathbf{B} = -\mu_0 \mathbf{J}, \quad k = \frac{\omega}{c}$$

得到. 对磁场的相对修正

$$\frac{\delta B}{B} \sim (kR)^2 = \left( \frac{\omega R}{c} \right)^2$$

对电场的相对修正

$$\frac{\delta E}{E} \sim (kR)^2$$

(b) 静磁场近似使用瞬时电流, 无时延效应. 当观察点距离至少与波长的量级相当时 (相位差  $\approx 1\text{rad}$ ), 重叠的波动效应不能忽略. 波长  $\lambda = 2\pi c/\omega$ . 失效临界位置  $z_{\text{break}} \sim \frac{c}{\omega} \approx \frac{\lambda}{2\pi}$ .

### 3 Problem 3

Consider a one-dimensional non-relativistic particle of mass  $m$  and kinetic energy  $E$  scattering off the potential  $U(x)$  composed of two static delta-functions

$$U(x) = \beta[\delta(x) + \delta(x - a)]$$

1. Can the particle tunnel through this barrier without reflection? Explain your answer.
2. If so, at what value(s) of the kinetic energy does this happen?
3. Now consider a three-dimensional non-relativistic particle of mass  $m$  and kinetic energy  $E$  scattering off the potential  $U(\vec{r})$  composed of two static delta-functions

$$U(\vec{r}) = \beta_3(\delta^{(3)}(\vec{r} - \vec{r}_1) + \delta^{(3)}(\vec{r} - \vec{r}_2)). \quad (5)$$

Suppose that for each of the delta-functions in the above equation the S-wave scattering length  $a < 0$ .

4. Write explicitly the S-wave bound wave-function near each of the centers. Check that each does not support a bound state for  $a < 0$ .
5. Can the potential  $U(\vec{r})$  with two centers support a bound state? If so, under what conditions? Explain your answer.

**Hint:** For a hard core potential of radius  $R$ , the S-wave scattering length  $a$  is defined as

$$\left. \frac{d \log \chi(R)}{dR} \right|_R = -1/a,$$

with  $\chi(r)$  the S-wave reduced wave-function  $\chi(r) \equiv r\psi(r)$ . If  $\psi(r)$  corresponds to a bound state, think about the constraints on the scattering length  $a$ .

#### 解答

1. 可以. 当入射波能量满足一定条件时, 两个  $\delta$  势之间的干涉效应可使反射波完全抵消, 实现完全透射. 这类似于光学中的增透膜或量子力学中的共振透射.
2. 利用转移矩阵方法. 单个  $\delta$  势的转移矩阵为

$$M_0 = \begin{pmatrix} 1 - i\eta & -i\eta \\ i\eta & 1 + i\eta \end{pmatrix}, \quad \eta = \frac{m\beta}{\hbar^2 k}$$

自由传播矩阵  $P(a) = \text{diag}(e^{ika}, e^{-ika})$ . 总转移矩阵  $M_{\text{total}} = M_0 P(a) M_0$ .

入射波  $\begin{pmatrix} 1 \\ r \end{pmatrix}$  与透射波  $\begin{pmatrix} t \\ 0 \end{pmatrix}$  满足

$$\begin{pmatrix} t \\ 0 \end{pmatrix} = M_{\text{total}} \begin{pmatrix} 1 \\ r \end{pmatrix}$$

无反射 ( $r = 0$ ) 要求  $(M_{\text{total}})_{21} = 0$ , 即

$$i\eta[(1 - i\eta)e^{ika} + (1 + i\eta)e^{-ika}] = 0$$

整理得

$$(1 - i\eta)e^{ika} + (1 + i\eta)e^{-ika} = 0 \implies e^{-2ika} = -\frac{1 - i\eta}{1 + i\eta}$$

右边模为 1, 可写为  $-e^{-2i \arctan \eta}$ , 故

$$-2ka = \pi - 2 \arctan \eta + 2n\pi \implies ka = \arctan \eta - \frac{\pi}{2} + n\pi$$

取正切得

$$\tan(ka) = -\cot(\arctan \eta) = -\frac{1}{\eta} = -\frac{\hbar^2 k}{m\beta}$$

此即无反射的能量条件, 存在一系列分立  $k_n$  使透射率  $T = 1$ .

3. 对于单个球对称  $\delta$  势,  $S$  波径向约化波函数  $\chi(r) = r\psi(r)$  在  $r > 0$  满足自由方程

$$-\frac{\hbar^2}{2m}\chi''(r) = E\chi(r)$$

考虑束缚态  $E = -\frac{\hbar^2 \kappa^2}{2m} < 0$ , 则  $\chi''(r) = \kappa^2 \chi(r)$ , 有界解为  $\chi(r) = Ce^{-\kappa r}$ . 在  $r \rightarrow 0$  处, 边界条件由散射长度  $a$  定义为

$$\left. \frac{d \ln \chi(r)}{dr} \right|_{r \rightarrow 0} = -\frac{1}{a}$$

代入  $\chi(r) = Ce^{-\kappa r}$  得

$$\frac{\chi'(0)}{\chi(0)} = -\kappa = -\frac{1}{a} \implies \kappa = \frac{1}{a}$$

若  $a < 0$ , 则  $\kappa < 0$ , 此时  $\chi(r) = Ce^{|\kappa|r}$  在  $r \rightarrow 0$  时趋于常数, 但在  $r > 0$  时指数增长, 不满足平方可积. 因此单个中心在  $a < 0$  时不能支持束缚态.

4. 可以. 两个中心相距  $R = |\vec{r}_1 - \vec{r}_2|$ , 总波函数用格林函数叠加

$$\psi(\vec{r}) = C_1 \frac{e^{-\kappa|\vec{r}-\vec{r}_1|}}{|\vec{r}-\vec{r}_1|} + C_2 \frac{e^{-\kappa|\vec{r}-\vec{r}_2|}}{|\vec{r}-\vec{r}_2|}$$

其中  $\kappa > 0$  待定, 束缚态能量  $E = -\hbar^2 \kappa^2 / (2m)$ . 在每个中心附近匹配边界条件.

以  $\vec{r}_1$  为中心, 令  $\rho = |\vec{r} - \vec{r}_1| \rightarrow 0$ , 则  $|\vec{r} - \vec{r}_2| \rightarrow R$ . 约化波函数

$$\chi_1(\rho) = \rho\psi \approx C_1(1 - \kappa\rho) + C_2\rho \frac{e^{-\kappa R}}{R}$$

其对数导数在  $\rho \rightarrow 0$  处

$$\frac{\chi_1'(0)}{\chi_1(0)} = \frac{-C_1\kappa + C_2e^{-\kappa R}/R}{C_1} = -\kappa + \frac{C_2}{C_1} \frac{e^{-\kappa R}}{R}$$

边界条件要求该值等于  $-1/a$ , 故

$$-\kappa + \frac{C_2}{C_1} \frac{e^{-\kappa R}}{R} = -\frac{1}{a} \implies \frac{C_2}{C_1} \frac{e^{-\kappa R}}{R} = \kappa - \frac{1}{a}$$

同理, 在  $\vec{r}_2$  处得

$$\frac{C_1}{C_2} \frac{e^{-\kappa R}}{R} = \kappa - \frac{1}{a}$$

两式相乘得

$$\left( \frac{e^{-\kappa R}}{R} \right)^2 = \left( \kappa - \frac{1}{a} \right)^2 \implies \kappa - \frac{1}{a} = \pm \frac{e^{-\kappa R}}{R}$$

由于  $a < 0$ ,  $1/a < 0$ ;  $\kappa > 0$ , 左边  $\kappa - 1/a > 0$ . 右边若取负号, 则  $\kappa - 1/a = -e^{-\kappa R}/R < 0$ , 矛盾. 故只能取正号

$$\kappa - \frac{1}{a} = \frac{e^{-\kappa R}}{R} \implies \frac{1}{a} = \kappa - \frac{e^{-\kappa R}}{R}$$

令  $f(\kappa) = \kappa - \frac{e^{-\kappa R}}{R}$ , 则  $f(0) = -\frac{1}{R} < 0$ ,  $f(+\infty) = +\infty$ , 且  $f'(\kappa) = 1 + e^{-\kappa R} > 0$ , 故  $f$  严格单调递增. 方程  $f(\kappa) = 1/a$  有正根  $\kappa$  当且仅当

$$\frac{1}{a} > f(0) = -\frac{1}{R} \implies -\frac{1}{a} < R$$

由于  $a < 0$ , 记  $|a| = -a$ , 条件化为

$$|a| > R$$

即两中心距离小于散射长度的绝对值时, 存在一个束缚态. 反之, 若  $R \geq |a|$ , 则无束缚态. 这一结果类似于两个吸引中心在足够近时形成分子轨道态, 即使每个孤立中心无束缚态.

## 4 Problem 4

1. Write down the condition for thermodynamic equilibrium between the liquid and gas phases along a liquid-gas coexistence curve.
2. Using your solution to part 1 and taking into account the entropy change at a liquid-gas phase transition, derive the relation for the vapor pressure along a liquid-gas coexistence curve.

解答

1. 沿液 - 气共存曲线, 热力学平衡要求温度  $T$ 、压强  $p$  相等且化学势相等, 即  $\mu_{\text{liquid}}(T, p) = \mu_{\text{gas}}(T, p)$ .
2. 化学势全微分为  $d\mu = -s dT + v dp$ ,  $s$  为摩尔熵,  $v$  为摩尔体积. 沿共存曲线  $d\mu_l = d\mu_g$ , 故  $-s_l dT + v_l dp = -s_g dT + v_g dp$ , 整理得  $\frac{dp}{dT} = \frac{s_g - s_l}{v_g - v_l}$ . 相变潜热  $L = T(s_g - s_l)$ , 代入得克拉佩龙方程  $\frac{dp}{dT} = \frac{L}{T(v_g - v_l)}$ . 若气体视为理想气体且  $v_g \gg v_l$ , 则  $v_g \approx RT/p$ , 有  $\frac{dp}{dT} = \frac{Lp}{RT^2}$ . 当  $L$  近似常数时积分得  $p = p_0 \exp(-L/(RT))$ , 即蒸气压力随温度指数上升.

## 5 Problem 5

Consider the  $\phi^4$  model with a real scalar field  $\phi(x)$  in  $3 + 1$  dimensional Minkowski spacetime with metric  $(-, +, +, +)$ . Its Lagrangian density is

$$L = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 - \frac{1}{4!}g\phi^4.$$

1. Write down the propagator and the interaction vertex for this model in momentum space.
2. Compute the one-loop correction to the 4-point function using dimensional regularization.
3. Determine the counterterm needed to cancel the divergence and write the renormalized coupling at 1-loop order (you may use minimal subtraction).
4. Write down the renormalization group flow equation for the coupling constant.

Useful formulas:

$$(AB)^{-1} = \int_0^1 dx [xA + (1-x)B]^{-2}$$

$$\int d^d k \frac{1}{(-k^2 - 2p \cdot k - M^2 + i\epsilon)^s} = (-1)^s i\pi^{d/2} \frac{\Gamma(s - d/2)}{\Gamma(s)} (-p^2 + M^2 - i\epsilon)^{d/2-s}$$

$$\Gamma(z) = 1/z - \gamma + O(z)$$

解答

1. 动量空间中实标量场的费曼传播子为  $\Delta(p) = i/(p^2 + m^2 - i\epsilon)$ , 其中  $p^2 = -p_0^2 + \mathbf{p}^2$ . 相互作用项  $-\frac{g}{4!}\phi^4$  给出的顶点因子为  $-ig$ .

2. 考虑  $s$  通道, 外动量  $p_1, p_2$  合成  $P = p_1 + p_2$ ,  $s = P^2$ , 对称因子  $1/2$ , 振幅为

$$i\mathcal{M}_s = \frac{g^2}{2} \int \frac{d^d k}{(2\pi)^d} \frac{1}{(k^2 + m^2)((P - k)^2 + m^2)} \equiv \frac{g^2}{2} I(s)$$

用费曼参数化  $\frac{1}{AB} = \int_0^1 dx [xA + (1-x)B]^{-2}$ , 令  $A = k^2 + m^2$ ,  $B = (P - k)^2 + m^2$ , 配方得分母  $(k - xP)^2 + \Delta$ , 其中  $\Delta = m^2 + x(1-x)s$ . 作平移  $l = k - xP$  得

$$I(s) = \int_0^1 dx \int \frac{d^d l}{(2\pi)^d} \frac{1}{(l^2 + \Delta)^2}$$

取  $d = 4 - 2\epsilon$ , 得

$$\int \frac{d^d l}{(2\pi)^d} \frac{1}{(l^2 + \Delta)^2} = \frac{i}{(4\pi)^{d/2}} \frac{\Gamma(2 - d/2)}{\Gamma(2)} \Delta^{d/2-2} = \frac{i}{16\pi^2} \left( \frac{1}{\epsilon} - \gamma + \ln(4\pi) - \ln \Delta + O(\epsilon) \right)$$

故

$$I(s) = \frac{i}{16\pi^2} \int_0^1 dx \left( \frac{1}{\epsilon} - \gamma + \ln(4\pi) - \ln(m^2 + x(1-x)s - i\epsilon) \right)$$

一圈总修正为  $\frac{g^2}{2}(I(s) + I(t) + I(u))$ , 发散部分为  $\frac{g^2}{2} \cdot 3 \cdot \frac{i}{16\pi^2\epsilon} = \frac{3ig^2}{32\pi^2\epsilon}$ .

3. 在最小减除 (MS) 方案中, 令裸耦合  $g_0 = g + \delta g$ , 要求抵消发散, 得  $\delta g = \frac{3g^2}{32\pi^2\epsilon}$ . 重整化后的耦合常数记为  $g(\mu)$ , 在 MS 方案下有限.
4.  $\beta$  函数定义为  $\beta(g) = \mu \frac{dg}{d\mu}$ . 由  $g_0$  与  $\mu$  无关可导出  $\beta(g) = \frac{3g^2}{16\pi^2}$ .

## 6 Problem 6

The Klein disk model is an analytical description of non-Euclidean geometry in terms of coordinates  $(x^1, x^2)$  for points inside a unit disk. The distance  $d(\vec{x}, \vec{y})$  between  $\vec{x} = (x^1, x^2)$  and  $\vec{y} = (y^1, y^2)$  (both inside the disk) is defined by

$$\cosh d(\vec{x}, \vec{y}) = \frac{1 - \vec{x} \cdot \vec{y}}{\sqrt{1 - \vec{x} \cdot \vec{x}} \sqrt{1 - \vec{y} \cdot \vec{y}}},$$

where  $\vec{x} \cdot \vec{y} = x^1 y^1 + x^2 y^2$ .

1. Compute the metric components  $g_{11}, g_{12}, g_{22}$  from the formula for  $d$  by considering  $\vec{y}$  near  $\vec{x}$  ( $y^j = x^j + dx^j$ ).
2. Introduce standard polar coordinates  $(r, \theta)$  and write the line element  $ds^2 = g_{\mu\nu} dx^\mu dx^\nu$  in polar coordinates.
3. Determine the nonvanishing Christoffel symbols in polar coordinates.
4. Compute the component  $R_{r\theta r\theta}$  of the Riemann tensor.
5. Show that for the Klein disk model, the Riemann curvature has the form  $R_{\mu\nu\rho\sigma} = K(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho})$  and determine  $K$ .
6. Compute the scalar curvature.

**解答**

1. 令  $\vec{y} = \vec{x} + d\vec{x}$ , 记  $r^2 = |\vec{x}|^2$ . 展开  $\cosh d \approx 1 + d^2/2$  并比较  $d\vec{x}$  的二阶项, 得线元  $ds^2 = \frac{d\vec{x}^2}{1 - r^2} + \frac{(\vec{x} \cdot d\vec{x})^2}{(1 - r^2)^2}$ . 由此读出  $g_{11} = \frac{1}{1 - r^2} + \frac{(x^1)^2}{(1 - r^2)^2}$ ,  $g_{22} = \frac{1}{1 - r^2} + \frac{(x^2)^2}{(1 - r^2)^2}$ ,  $g_{12} = g_{21} = \frac{x^1 x^2}{(1 - r^2)^2}$ .
2. 设  $x^1 = r \cos \theta$ ,  $x^2 = r \sin \theta$ , 则  $d\vec{x}^2 = dr^2 + r^2 d\theta^2$ ,  $\vec{x} \cdot d\vec{x} = r dr$ . 代入得  $ds^2 = \frac{dr^2}{1 - r^2} + \frac{r^2 d\theta^2}{1 - r^2} + \frac{r^2 dr^2}{(1 - r^2)^2} = \frac{dr^2}{(1 - r^2)^2} + \frac{r^2}{1 - r^2} d\theta^2$ . 故  $g_{rr} = 1/(1 - r^2)^2$ ,  $g_{\theta\theta} = r^2/(1 - r^2)$ ,  $g_{r\theta} = 0$ .

3. 计算得  $\Gamma_{rr}^r = \frac{2r}{1-r^2}$ ,  $\Gamma_{\theta\theta}^r = -r$ ,  $\Gamma_{r\theta}^\theta = \Gamma_{\theta r}^\theta = \frac{1}{r(1-r^2)}$ , 其余为零.
4. 黎曼张量分量  $R_{r\theta r\theta} = g_{rr}R_{\theta r\theta}^r$ . 计算  $R_{\theta r\theta}^r = \partial_r\Gamma_{\theta\theta}^r - \partial_\theta\Gamma_{\theta r}^r + \Gamma_{r\lambda}^r\Gamma_{\theta\theta}^\lambda - \Gamma_{\theta\lambda}^r\Gamma_{r\theta}^\lambda$ . 代入已知符号得  $R_{\theta r\theta}^r = -\frac{r^2}{1-r^2}$ , 故  $R_{r\theta r\theta} = \frac{1}{(1-r^2)^2} \cdot \left(-\frac{r^2}{1-r^2}\right) = -\frac{r^2}{(1-r^2)^3}$ .
5. 二维流形中黎曼张量可写为  $R_{\mu\nu\rho\sigma} = K(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho})$ . 取分量  $R_{r\theta r\theta} = K(g_{rr}g_{\theta\theta} - g_{r\theta}^2) = Kg_{rr}g_{\theta\theta}$ . 代入  $g_{rr}g_{\theta\theta} = \frac{1}{(1-r^2)^2} \cdot \frac{r^2}{1-r^2} = \frac{r^2}{(1-r^2)^3}$ , 与第4问结果比较得  $K = -1$ .
6. 由  $R_{\mu\nu} = g^{\rho\sigma}R_{\mu\rho\nu\sigma} = Kg_{\mu\nu}(2-1) = Kg_{\mu\nu}$ , 再缩并得  $R = g^{\mu\nu}R_{\mu\nu} = 2K = -2$ . 故标量曲率为  $-2$ .